

UTILITY USAGE PREDICTION TOOL

AI/ML Internship Project Report

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I sincerely thank CODEVEDX for providing the opportunity to develop this project. This internship helped me strengthen my knowledge of Python, CSV file handling, and basic Machine Learning.

Abstract

The Utility Usage Prediction Tool is a console-based Python application that stores electricity usage records in a CSV file and predicts future electricity consumption using the Linear Regression algorithm. The project combines file handling, data management, and machine learning in a simple application.

Objectives

- Develop a menu-driven application.
- Manage electricity records using CSV files.
- Perform CRUD operations.
- Predict future electricity usage using Linear Regression.
- Improve Python programming skills.

Problem Statement

Manual maintenance of electricity usage records is time-consuming and does not provide prediction capabilities. This project offers a simple digital solution to manage records and estimate future consumption.

Existing System

Traditional record keeping using notebooks or spreadsheets is difficult to maintain, update, and analyze.

Proposed System

The proposed system stores records in a CSV file, allows users to add, update, delete, and view data, and predicts future electricity usage using historical records.

Software Requirements

Component	Requirement
Operating System	Windows 10/11
Language	Python 3.x
IDE	Visual Studio Code
Libraries	Pandas, Scikit-learn

Technologies Used

- Python
- Pandas
- Scikit-learn
- CSV Files
- Linear Regression

Project Workflow

1. Load or create CSV file
2. Display menu
3. Perform CRUD operations
4. Train Linear Regression model
5. Predict future usage
6. Save updated records

Dataset

Dataset File: Project1.csv

Columns:

- Day
- Electricity_Usage

Features

- Menu-driven console
- CSV file handling
- Add, Update, Delete and View records

- Prediction using Machine Learning
- Exception handling

Conclusion

The project successfully demonstrates the practical use of Python programming, CSV file handling, and Machine Learning. It provides an efficient way to manage electricity usage records and estimate future consumption.

References

- [Python Documentation](#)
- [Pandas Documentation](#)
- [Scikit-learn Documentation](#)